

Symplectic Lagrangian Phantom

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Summary

Pass 79 asks whether ϵ itself is a symplectic object. The corrected answer is subtler: ϵ has a unique nonzero derived self-pairing, but that pairing is degree-shifted, alternating, and degenerate. The right nondegenerate finite picture is a hyperbolic pair with complementary Lagrangians ϵ and $\mathbb{Q}$.

Main Claims

- **No intrinsic Darboux form.** The candidate pairing $\epsilon \otimes \epsilon \rightarrow \mathbb{Z}[m]$ is nonzero only for $m = 2$.
- **Finite-adele generator.** The extension $0 \rightarrow \mathbb{Q} \rightarrow \mathbb{A}_f \rightarrow \epsilon \rightarrow 0$ generates $\mathrm{Ext}_{\mathrm{Solid}}^1(\epsilon, \mathbb{Q})$.
- **Alternating but degenerate.** The odd degree gives the swap sign -1 , so the pairing is alternating. Its adjoint is not an isomorphism.
- **Hyperbolic correction.** Finite pairings $\mathbb{Z}/N \times \mathbb{Z}/N \rightarrow \mathbb{Q}/\mathbb{Z}$ are perfect; the symplectic object should be read as a hyperbolic plane with two complementary Lagrangians.

Machine Verification

`code/scripts/check-pass79.py` produced `artifacts/reports/pass79-symplectic-lagrangian-phantom-check.json` with overall PASS. It checks vanishing of $\mathrm{RHom}(\mathrm{Zhat}, \mathbb{Q})$, the degree table for self-pairings, the finite-adele extension class, the alternating sign, and the Darboux/projector no-go.

Next Use

This pass sets up the later metaplectic question: if ϵ and $\mathbb{Q}$ are complementary Lagrangians, the decisive test is whether the Weyl flip $\epsilon \rightarrow \mathbb{Q}$ survives in the solid limit.